

Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	18 February 2026
Team ID	LTVIP2026TMIDS54574
Project Name	Smart Sorting: Transfer Learning for Identifying rotten fruits and vegetables
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

product backlog and sprint schedule

	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Preprocessing	USN-1	As a developer, I want to collect images of fresh and rotten fruits and vegetables.	2	High	K Vinodh
Sprint-1		USN-2	As a developer, I want to preprocess the dataset for training (resize, normalize, augment).	1	High	K Vinodh
Sprint-1		USN-3	As a developer, I want to resize and normalize the images.	2	High	K Vinodh
Sprint-1		USN-4	As a developer, I want to split images into training and test sets.	1	High	K Vinodh
Sprint-1		USN-5	As a developer, I want to apply data augmentation (flip, rotate, etc.).	2	High	K Vinodh

Sprint-1		USN-6	As a developer, I want to perform label encoding for classification.	2	High	K Vinodh
Sprint-2	Model Training & Evaluation	USN-7	As a developer, I want to load the pretrained VGG16 model.	2	High	K Ujwala Sai Sree
Sprint-2		USN-8	As a developer, I want to modify the final layers for classification.	2	Medium	K Ujwala Sai Sree
Sprint-2		USN-9	As a developer, I want to train the model on the dataset.	3	High	K Ujwala Sai Sree
Sprint-2		USN-10	As a developer, I want to save the trained model.	1	Medium	K Ujwala Sai Sree
Sprint-2		USN-11	As a developer, I want to test the model on unseen data.	2	High	K Kusuma Kumari
Sprint-2		USN-12	As a developer, I want to generate accuracy, precision, and recall metrics.	2	Medium	K Kusuma Kumari
Sprint-3	Deployment & Application Interface	USN-13	As a developer, I want to create an HTML page for image upload.	2	Medium	K Kusuma Kumari
Sprint-3		USN-14	As a developer, I want to display prediction results.	2	Medium	K Kusuma Kumari

Sprint-3		USN-15	As a developer, I want to develop a Flask backend.	3	High	K Uma Bharathi
Sprint-3		USN-16	As a developer, I want to connect the frontend to the backend.	2	Medium	K Uma Bharathi
Sprint-3		USN-17	As a developer, I want to setup and test the final application.	2	High	K Uma Bharathi

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	5 Days	2 February 2026	6 February 2026	10	6 February 2026
Sprint-2	12	5 Days	7 February 2026	11 February 2026	12	11 February 2026
Sprint-3	13	5 Days	12 February 2026	16 February 2026	13	16 February 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\textit{sprint duration}}{\textit{velocity}} = \frac{20}{10} = 2$$

Velocity

- Velocity = Total Story Points Completed / Number of Sprints
- Total Story Points = 10 + 12 + 13 = **35**
- Number of Sprints = 3
- **Velocity = 35 / 3 = 11.67 ≈ 12 (Story Points per Sprint)**